Optimization of Machining Parameters in Turning of Aluminium Alloy 6061 Using Taguchi Method

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Abstract

This paper studies the effect of varying machining parameters in turning on surface roughness (R_a and R_z), roundness (Ø) and material removal rate (MRR) for Aluminium Alloy 6061 using uncoated carbide insert under dry cutting conditions. CNC turning machine was employed to conduct experiments on aluminium alloy 6061. The cutting parameters used are cutting speed, feed rate and depth of cut. The effect of cutting parameters on surface roughness (R_a and R_z), roundness (Ø) and MRR is determined. Experiments have been conducted using Taguchi's experimental design technique. In this work, Taguchi technique with an L9 orthogonal array has been used to find the influence of cutting parameters on surface roughness (R_a and R_z), roundness and MRR. Using S/N ratio method, ranking of the cutting parameters are done and it is observed that for surface roughness (R_a), surface roughness (R_z), roundness (Ø) and MRR, feed rate is the most influencing parameter followed by depth of cut and by the cutting speed. The optimal cutting conditions for surface roughness (R_a & R_z) and Roundness (Ø) are arrived at the cutting speed of 165.32 m/min, feed rate of 0.05 mm and depth of cut of 0.2 mm. The optimal cutting conditions of MRR are arrived at the cutting speed of 165.32 m/min, feed rate of 0.1 mm/rev and depth of cut of 0.2 mm. The results are validated by analysis of variance method (ANOVA) and the percentage contribution of cutting speed, feed rate and depth of cut are determined.

Keywords: Taguchi's method, orthogonal array, S/N ratio, Surface Roughness, Optimization

1.0 INTRODUCTION

Turning is an important and widely used machining processes in engineering industries. In turning, parameters such as cutting speed, feed rate and depth of cut affects surface finish, roundness and MRR. Surface quality is an important performance characteristic to evaluate the productivity of machine tools as well as machined components [1]. Surface roughness is the critical quality indicator for machined surfaces [2]. Generally, the suitable cutting parameters are determined based on experience or by use of a handbook which does not guarantee optimal performance [3]. In order to achieve increased productivity, it is desirable to have maximum value of MRR. Many studies have been conducted to determine the effect of machining parameters on surface roughness, but only few studies have been conducted to determine the effect of machining parameters on roundness and MRR. A good quality turning surface can lead to improvement in strength properties such as fatigue strength, corrosion resistance and thermal resistance.

C. Ahilan et al. [4] investigated the effect of machining parameters on AISI 304 stainless steel using carbide tool insert and have designed experiments based on Taguchi's Design of Experiments. It was observed that feed rate is the dominant factor in determining surface roughness. R. Suresh et al. [5] have indicated that feed rate has the highest influence on the surface roughness followed by cutting speed and depth of cut while machining AISI4340 steel using multilayer coated carbide tool. S. Ramesh et al. [6] used response surface model for predicting surface roughness values in turning the aerospace titanium alloy and observed that feed rate was the most influencing factor followed by depth of cut and cutting speed. N. Muthukrishnan et al. [7] studied the effect of cutting parameters on surface roughness while turning Al/SiC-MMC using PCD insert with ANOVA and ANN analysis and observed that the feed rate has highest physical as well as statistical influence on the surface roughness right after the depth of cut and the cutting speed. Chorng-Jyh Tzeng et al. [8] have studied the effect of cutting parameters on roughness average, roughness maximum and roundness while turning SKD11 tool steel via grey relational analysis and identified depth of cut to be the most influential factor on the roughness average and the cutting speed is most influential factor on roughness maximum and the roundness. M. Vijaya Kini et al. [9] have investigated the effect of varying machining parameters in turning of GFRP on surface roughness and MRR and have concluded that for the surface roughness, the feed rate is the main influencing factor followed by the depth of cut and for MRR , the depth of cut is the influential factor followed by tool nose radius. In the present work, the effect of cutting parameters such as cutting speed, feed rate and depth of cut on surface roughness (R_a & R_z), roundness (Ø) and MRR while turning aluminium alloy 6061 has been determined from experiments conducted using Taguchi's experimental design technique.

2.0 MATERIAL TESTING AND SPECIFICATIONS

Chemical composition has been obtained using an Optical Emission Spectroscopy and is given in Table 1. This alloy has got good toughness, good weldability, good workability and excellent corrosion resistance as key properties. This alloy is widely used in the manufacturing of aircraft components, valves, bicycle frames, marine fittings and drive shafts. Specifications of the CNC machine used in the tests are given in Table 2. Aluminium alloy 6061 material rod of 24 mm diameter and 60 mm length was used in all the experiments. The hardness number of specimen is BHN33. The tool used is uncoated carbide insert tool. The specification of cutting tool is DCGT 11 T3 04. The specification of the shank is SDSCR 12 12 11 F3. The surface roughness was measured using the Surfcorder SE 1200, Surface profilometer with the following specifications. Standard: JIS 94, cut off: 0.8 mm, filter: Gauss, sampling length : 0.8 mm, evaluation length 4.00 mm, leveling ALL and measuring speed: 0.5 mm/s. The machining was done under dry cutting condition.

Table 1 Chemical Composition of Aluminium Alloy 6061

Sl.No.	Element	Wt %
1	Si	0.607%
2	Fe	0.120%
3	Cu	0.183%
4	Mn	0.084%
5	Mg	0.972%
6	Cr	0.077%
7	Zn	0.004%
8	Ti	0.029%
9	Al	97.9%

Table 2: Specifications of the CNC Lathe

Sl. No.	Description	Specification
1	Controller	SIEMENS 802D SL
2	Swing over bed	275 mm
3	Swing over cross slide	80 mm
4	Maximum turning diameter/length	80 mm/180 mm
5	Chuck size	100 mm
6	X axis travel	95 mm (190 mm in dia)
7	Z axis travel	210 mm
8	Positioning accuracy	0.010
9	Spindle motor capacity	3.73 kW

3.0 EXPERIMENTAL WORK

Taguchi's L9 orthogonal array was used to design the experiments with three factors and three levels. Experiments were conducted based on the Taguchi's method which is a powerful tool used in design of experiments. [10]. Taguchi advocates use of orthogonal array designs to assign the factors chosen for the experiment. The advantage of Taguchi method is that it uses a special design of orthogonal arrays to study the entire parameter space with only a small number of experiments. Compared to the conventional approach of experimentation, this method reduces drastically the number of experiments that are required to model the response functions [11]. The assignment of the levels to the factors and the various parameters used are given in Table 3.

Experiments were conducted on a CNC lathe and the surface roughness (R_a and R_z), roundness and the material removal rate (MRR) have been determined. The experimental results for L9 orthogonal array with their corresponding calculated S/N ratios are given in Table 4. (a) and (b). Statistical analysis was performed with the help of MINITAB software for obtaining main effects. For optimization of process parameters, the experimental results are transferred to Signal-to-Noise (S/N, dB) ratio to measure the quality characteristics deviating from the desired values. Signal-to-Noise ratio for surface roughness (R_a & R_z), and roundness are calculated as per smaller-the-better characteristics proposed by Taguchi [12] and is expressed as

$$S/N_{SB} = -10 \log_{10}\left(\frac{1}{n}\sum_{i=1}^{n} y_i{}^2\right) \tag{1}$$

Where y_i is the value of surface roughness for the i^{th} test in that trial.

Similarly, the S/N ratio for the Material Removal Rate (MRR) has also been calculated with consideration of larger-the-better characteristics i.e.

$$S/N_{SB} = -10 \log_{10}\left(\frac{1}{n}\sum_{i=1}^{n} \frac{1}{y_i{}^2}\right) \tag{2}$$

Where y_i is the value of material removal rate (MRR) for the i^{th} test in that trial.

Since the experimental design is orthogonal, it is then possible to separate out the effect of each cutting parameter at different levels. For example, the mean S/N ratio for the feed rate at levels 1,2 and 3 can be calculated by averaging the S/N ratios for the experiments 1-3, 4-6 and 7-9 respectively. The mean S/N ratio for each level of other cutting parameters can be computed in the similar manner.

4.0 RESULTS AND DISCUSSION

4.1 Effect of Cutting Parameters on Surface Roughness (R_a)

Fig. 1. and Fig. 2 shows the plots of main effects for S/N ratio and the interaction plot for surface roughness (R_a).

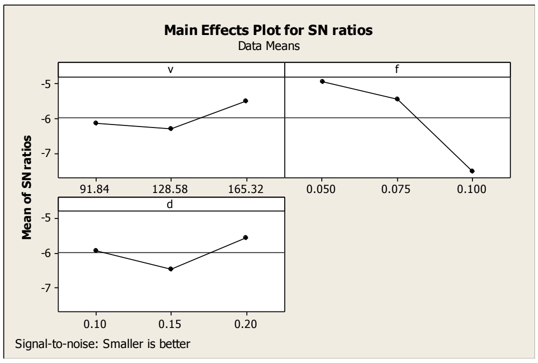


Fig. 1: Main effects of S/N Ratio for R_a

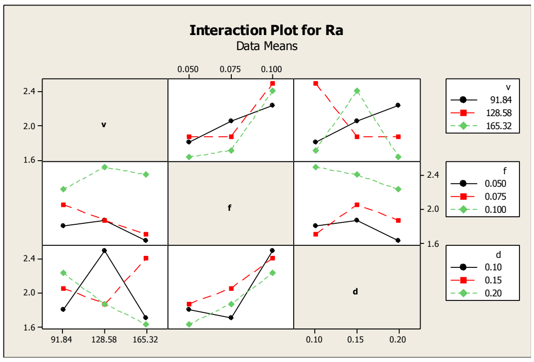


Fig. 2: Interaction Plot for R_a

From the main effect plot of S/N ratio we observed that the order of process parameters that influences the surface roughness (R_a) are feed followed by depth of cut and cutting speed. Optimal results could be found out from the main effect plot selecting the highest levels of S/N ratio values from the response table of S/N ratios for R_a. The response for S/N ratio of R_a is shown in Table 5. In the Table 5, the level with greater S/N ratios are marked in each column and the factor are ranked based on the corresponding Delta values .Therefore, based on the S/N analysis, the optimal process parameters for surface roughness (R_a) are determined as follows: cutting speed at level 3 (165.32 m/min), feed rate at level 1 (0.05 mm/rev) and depth of cut at level 3 (0.2mm). The feed rate is found to be most influencing factor affecting surface roughness followed by depth of cut and cutting speed. The ANOVA for R_a is given in Table 6. It is clear from the ANOVA table that feed rate is the most significant factor for surface roughness (R_a). Among the three cutting parameters, the contribution of feed rate in influencing the surface roughness (R_a) is 83.1% followed by depth of cut whose contribution is 7.9% and that of cutting speed is 5.76%. The contribution of depth of cut and cutting speed are insignificant.

Table 3: Assignment of the Levels to the Factors

Sl.No	Factors	Unit	Symbols	Level 1	Level 2	Level 3
1	Cutting speed	m/min	v	91.84	128.58	165.32
2	Feed rate	mm/rev	f	0.05	0.075	0.1
3	Depth of cut	mm	d	0.1	0.15	0.2

Table 4. (a). Experimental Results for L9 Orthogonal Array

Trial No.	Actual value			R_a (μm)	S/N ratio for R_a	R_z (μm)	S/N ratio for R_z
	v (m/min)	f (mm/rev)	d (mm)				
1	91.84	0.05	0.1	1.81	-5.15357	5.26	-14.4197
2	91.84	0.075	0.15	2.058	-6.26891	5.98	-15.534
3	91.84	0.1	0.2	2.235	-6.98555	6.497	-16.2543
4	128.58	0.05	0.15	1.876	-5.46466	5.453	-14.7327
5	128.58	0.075	0.2	1.876	-5.46466	5.552	-14.889
6	128.58	0.1	0.1	2.502	-7.96575	7.272	-17.2331
7	165.32	0.05	0.2	1.632	-4.2544	4.76	-13.5521
8	165.32	0.075	0.1	1.711	-4.665	4.972	-13.9306
9	165.32	0.1	0.15	2.41	-7.64034	7.003	-16.9057

Table 4. (b). Experimental Results for L9 Orthogonal Array

Trial No	Actual value			Roundness (Ø)	S/N ratio for Roundness (Ø)	MRR (cm^3/min)	S/N ratio for MRR
	v (m/min)	f (mm/rev)	d (mm)				
1	91.84	0.05	0.1	3.28	-10.3175	0.143603	-16.8567
2	91.84	0.075	0.15	3.683	-11.324	0.27507	-11.2111
3	91.84	0.1	0.2	4.018	-12.0802	0.470602	-6.5469
4	128.58	0.05	0.15	3.358	-10.5216	0.27507	-11.2111
5	128.58	0.075	0.2	3.368	-10.5474	0.49443	-6.1179
6	128.58	0.1	0.1	4.482	-13.0294	0.375577	-8.506
7	165.32	0.05	0.2	2.936	-9.3551	0.429231	-7.3462
8	165.32	0.075	0.1	3.068	-9.7371	0.331017	-9.603
9	165.32	0.1	0.15	4.314	-12.6976	0.640328	-3.872

4.1.1 Predicting the Optimum Performance

After the optimal level has been selected, one could predict the optimum response using the following equation:

$$\gamma_{\text{predicted}} = \gamma_m + \sum_{i=1}^{n}(\gamma i - \gamma m) \qquad (3)$$

where γ_m is the total mean S/N ratio, γi is the mean S/N ratio at optimal level, n is the number of main design parameters that affect the quality characteristics. It is very essential to perform a confirmation experiment for the parameter design, particularly when less number of data is utilized for optimization. The purpose of this confirmation experiment is to verify the improvement in the quality characteristics. Based on the eq. (1) and (3), the surface roughness (R_a) is predicted for the optimal combination of parameters (v3-f1-d3). Lastly confirmation test was conducted using the optimum combination of parameters (v3-f1-d3). Table 7 shows the comparison of predicted surface roughness (R_a) with the actual surface roughness (R_a) using the optimal cutting parameters, and good agreement between the predicted and actual surface roughness (R_a) is being observed. The increase of the S/N ratio from the initial cutting parameters to optimal cutting parameters is 1.9996 dB.

Table 5: Response Table for Signal to Noise Ratios of R_a, Smaller is Better

Level	v (m/min)	f (mm/rev)	d (mm)
1	-6.136	-4.958	-5.928
2	-6.298	-5.466	-6.458
3	-5.52	-7.531	-5.568
Delta	0.778	2.573	0.89
Rank	3	1	2

Total mean S/N ratio = -5.984

Table 6: Results of ANOVA for Surface Roughness (R_a)

Source	DF	Seq SS	Adj SS	Adj MS	F	P	%
v	2	0.04403	0.04403	0.02202	1.79	0.358	5.76
f	2	0.63424	0.63424	0.31712	25.81	0.037	83.1
d	2	0.06029	0.06029	0.03015	2.45	0.29	7.9
Error	2	0.02458	0.02458	0.01229			3.22
Total	8	0.76315					

S = 0.110853   R-Sq = 96.78%   R-Sq(adj) = 87.12%

Table 7: Results of Confirmation Experiment for R_a

	Initial cutting parameters	Optimal cutting parameters	
		Prediction	Experiment
Level	v2f2d2	v3f1d3	v3f1d3
Surface Roughness (μm)	2.0544	1.599	1.632
SN Ratio	-6.254	-4.078	-4.2544

Improvement of SN ratio = 1.9996 dB

4.2 Effect of cutting parameters on Surface Roughness (R_z)

Fig. 3. and Fig. 4. shows the plots of main effects for S/N ratio and the interaction plot for surface roughness (R_z). From the main effect plot of S/N ratio we observed that the order of process parameters that influences the surface roughness (R_z) are feed followed by depth of cut and cutting speed. Optimal results could be found out from the main effect plot selecting the highest levels of S/N ratio values from the response table of S/N ratios for R_z. The response for S/N ratio of R_z is shown in Table 8. In the Table 8, the level with greater S/N ratios are marked in each column and the factor are ranked based on the corresponding delta values .Therefore, based on the S/N analysis, the optimal process parameters for surface roughness (R_z) are determined as follows: cutting speed at level 3 (165.32 m/min), feed rate at level 1 (0.05 mm/rev) and depth of cut at level 3 (0.2mm). The feed rate is found to be most influencing factor affecting surface roughness (R_z) followed by depth of cut and cutting speed. The ANOVA for R_z is given in Table 9. It is clear from the ANOVA table that feed rate is the most significant factor for surface roughness (R_z). Among the three cutting parameters, the contribution of feed rate in influencing the surface roughness (R_z) is 83.0% followed by depth of cut whose contribution is 7.0% and that of cutting speed is 6.43%. The contribution of depth of cut and cutting speed are insignificant.

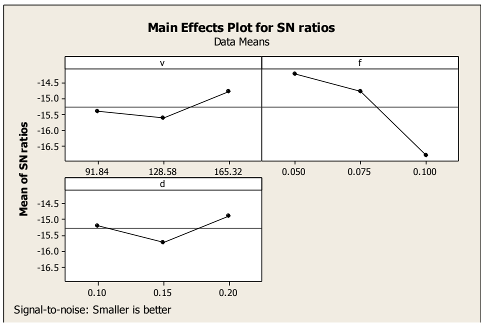


Fig. 3: Main Effects of S/N ratio for R_z

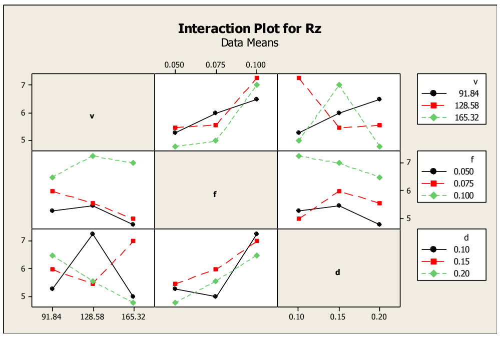


Fig. 4: Interaction plot for R_z

4.2.1 Predicting the optimum performance

Based on the eq. (1) and (3), the surface roughness (R_z) is predicted for the optimal combination of parameters (v3-f1-d3). Lastly confirmation test was conducted using the optimum combination of parameters (v3-f1-d3). Table 10 shows the comparison of predicted surface roughness with the actual surface roughness (R_z) using the optimal cutting parameters and good agreement between the predicted and actual surface roughness (R_z) is being observed. The increase of the S/N ratio from the initial cutting parameters to optimal cutting parameters is 2.028 dB.

Table 8: Response Table for Signal to Noise Ratios of R_z, Smaller is better

Level	v (m/min)	f (mm/rev)	d (mm)
1	-15.4	-14.23	-15.19
2	-15.62	-14.78	-15.72
3	-14.8	-16.8	-14.9
Delta	0.82	2.56	0.83
Rank	3	1	2

Total mean S/N ratio = -15.27

Table 9: Results of ANOVA for Surface Roughness (Rz)

Source	DF	Seq SS	Adj SS	Adj MS	F	P	%
v	2	0.4082	0.4082	0.2041	1.81	0.355	6.43
f	2	5.2620	5.262	2.6310	23.39	0.041	83
d	2	0.4443	0.4443	0.2222	1.98	0.336	7
Error	2	0.2249	0.2249	0.1125			3.54
Total	8	6.3394					

S = 0.335357   R-Sq = 96.45%   R-Sq(adj) = 85.81%

Table 10: Results of Confirmation Experiment for R_z

	Initial cutting parameters	Optimal cutting parameters	
		Prediction	Experiment
Level	v2f2d2	v3f1d3	v3f1d3
Surface Roughness(μm)	6.011	4.671	4.76
SN Ratio	-15.58	-13.39	-13.552

Improvement of SN ratio = 2.028 dB

4.3 Effect of cutting parameters on roundness (Ø)

Fig. 5 and Fig. 6. shows the plots of main effects for S/N ratio and the interaction plot for roundness (Ø). From the main effect plot of S/N ratio we observed that the order of process parameters that influences the roundness (Ø) are feed followed by depth of cut and cutting speed. Optimal results could be found out from the main effect plot selecting the highest levels of S/N ratio values from the response table of S/N ratios for roundness (Ø). The response for S/N ratio of roundness (Ø) is shown in Table 11. In the Table 11, the level with greater S/N ratios are marked in each column and the factor are ranked based on the corresponding delta values .Therefore, based on the S/N analysis, the optimal process parameters for roundness (Ø) are determined as follows: cutting speed at level 3 (165.32 m/min), feed rate at level 1 (0.05 mm/rev) and depth of cut at level 3 (0.2mm). The feed rate is found to be most influencing factor affecting roundness (Ø) followed by depth of cut and cutting speed. The ANOVA for roundness (Ø) is given in Table 12. It is clear from the ANOVA table that feed rate is the most significant factor for roundness. Among the three cutting parameters, the contribution of feed rate in influencing the roundness is 83.36% followed by depth of cut whose contribution is 7.38% and that of cutting speed is 5.92%. The contribution of depth of cut and cutting speed are insignificant.

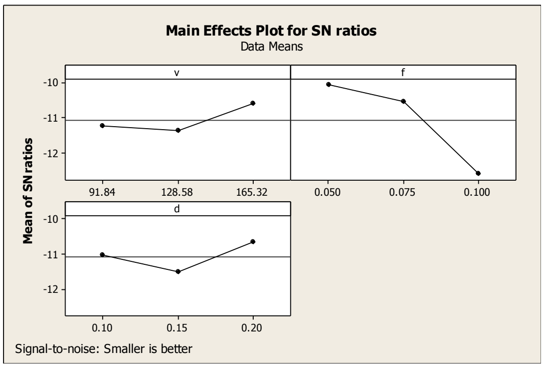


Fig. 5: Main Effects of S/N ratio for Roundness (Ø)

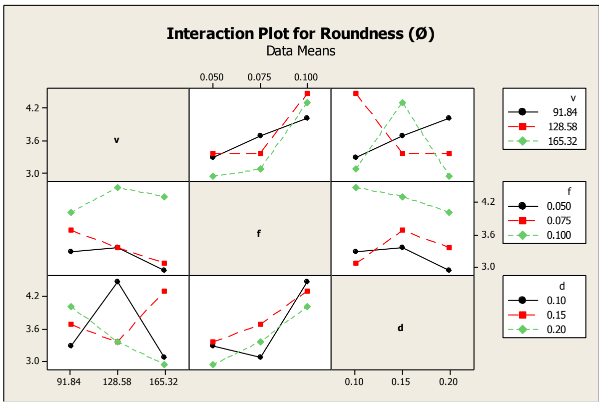


Fig. 6: Interaction Plot for Roundness (Ø)

4.3.1 Predicting the optimum performance

Based on the eq. (1) and (3), the roundness is predicted for the optimal combination of parameters (v3-f1-d3). Lastly confirmation test was conducted using the optimum combination of parameters (v3-f1-d3) . Table 13 shows the comparison of predicted roundness with the actual roundness using the optimal cutting parameters, and good agreement between the predicted and actual roundness is being observed. The increase of the S/N ratio from the initial cutting parameters to optimal cutting parameters is 1.931 dB.

Table 11: Response Table for Signal to Noise Ratios of Roundness (Ø), Smaller is Better

Level	v (m/min)	f (mm/rev)	d (mm)
1	-11.2	-10.06	-11.03
2	-11.4	-10.54	-11.51
3	-10.6	-12.6	-10.66
Delta	0.77	2.54	0.85
Rank	3	1	2

Total mean S/N ratio = -11.067

Table 12: Results of ANOVA for Roundness (Ø)

Source	DF	Seq SS	Adj SS	Adj MS	F	P	%
v	2	0.14258	0.14258	0.07129	1.78	0.359	5.92
f	2	2.00641	2.00641	1.0032	25.1	0.038	83.36
d	2	0.17786	0.17786	0.08893	2.23	0.31	7.38
Error	2	0.07992	0.07992	0.03996			3.32
Total	8	2.40677					

S = 0.199902   R-Sq = 96.68%   R-Sq(adj) = 86.72%

Table 13: Results of Confirmation Experiment for Roundness (Ø)

	Initial Cutting Parameters	Optimal Cutting Parameters	
		Prediction	Experiment
Level	v2f2d2	v3f1d3	v3f1d3
Roundness (Ø)	3.666	2.879	2.936
SN Ratio	-11.286	-9.186	-9.355

Improvement of SN Ratio = 1.931 dB

4.4. Effect of cutting parameters on Material Removal Rate (MRR)

In the similar way, the effect of cutting parameters on Material Removal rate (MRR) is analyzed. Fig. 7. shows main effects plots of S/N ratio for MRR and Fig. 8. is the graph showing interaction plot for MRR. From the main effect plot of S/N ratio we observed that the order of process parameters that influences the MRR are feed rate followed by depth of cut and cutting speed. Optimal results could be found out from the main effect plot selecting the highest levels of S/N ratio values from the response table of S/N ratios for MRR. The response of S/N ratio for MRR is shown in Table 14. In the table 14, the level with greater S/N ratios are marked in each column and the factor are ranked based on the corresponding delta values .Therefore, based on the S/N analysis, the optimal process parameters for MRR are determined as follows: cutting speed at level 3 (160.45 m/min), feed rate at level 3 (0.10 mm/rev) and depth of cut at level 3 (0.2mm). The feed rate is found to be most influencing factor affecting MRR followed by depth of cut and cutting speed. The ANOVA for MRR is given in Table 15. Among the three cutting parameters, the contribution of feed rate in influencing the MRR is 40.13% followed by depth of cut whose contribution is 29.31% and that of cutting speed is 25.36%. The contribution of feed rate, depth of cut and cutting speed are insignificant.

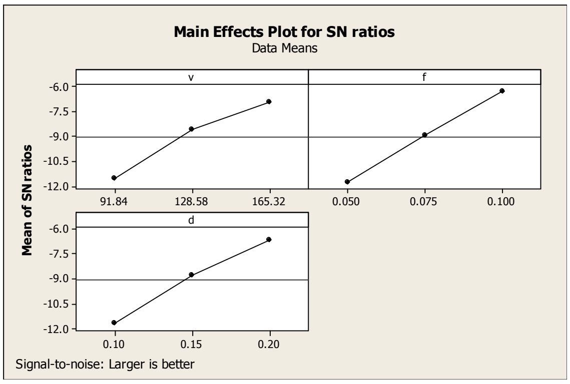
Table 14: Response Table for S/N Ratios for MRR, Larger is Better

Level	v (m/min)	f (mm/rev)	d (mm)
1	-11.5	-11.805	-11.655
2	-8.61	-8.977	-8.765
3	-6.94	-6.308	-6.67
Delta	4.598	5.496	4.985
Rank	3	1	2

Total mean S/N ratio = -9.03

4.4.1 Effect of Cutting Parameters on MRR

Based on the eq. (2) and (3), MRR is predicted for the optimal combination of parameters (v3-f3-d3). Lastly confirmation test was conducted using the optimum combination of parameters (v3-f3-d3). Table 16 shows the comparison of predicted MRR with the actual MRR using the optimal cutting parameters, and good agreement between the predicted and actual MRR is being observed. The increase of the SN ratio from the initial cutting parameters to optimal cutting parameters is 6.2268 dB.



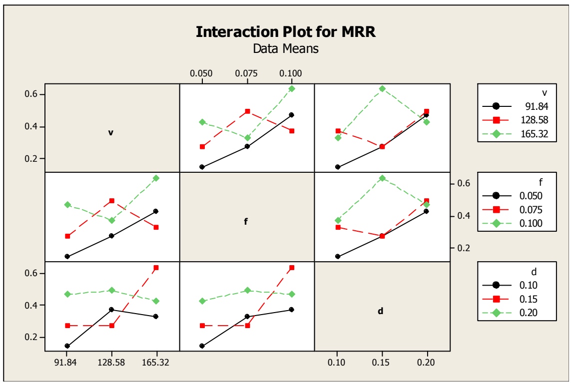


Fig. 7: Main Effects of S/N Ratio for MRR

Fig. 8: Interaction Plot for MRR

Table 15: Analysis of Variance for MRR

Source	DF	Seq SS	Adj SS	Adj MS	F	P	%
v	2	0.043571	0.043571	0.021786	4.9	0.17	25.36
f	2	0.068957	0.068957	0.034479	7.75	0.114	40.13
d	2	0.05037	0.05037	0.025185	5.66	0.15	29.31
Error	2	0.008897	0.008897	0.004449			5.1
Total	8	0.171796					

S = 0.0666988 R-Sq = 94.82% R-Sq(adj) = 79.28%

Table 16: Results of Confirmation Experiment for MRR

	Initial cutting parameters	Optimal cutting parameters	
		Prediction	Experiment
Level	V2f2d2	V3f3d3	V3f3d3
MRR (cm3/min)	0.38485	0.80742	0.7882
SN Ratio	-8.294	-1.858	-2.0672

Improvement of SN Ratio = 6.2268 dB

5.0 CONCLUSIONS

From the analysis of the surface roughness and MRR during the turning of the aluminium alloy 6061 using an uncoated carbide insert, the following conclusions were drawn:

- Basically, surface roughness, roundness and material removal rate are strongly correlated with cutting parameters such as cutting speed, feed rate and depth of cut. Hence, optimization of the cutting parameters based on the parameter design of the Taguchi method is adopted in this chapter.
- As per the results of the experiment, the feed rate was found to influence the surface roughness (R_a and R_z), roundness and MRR more significantly than the cutting speed and depth of cut, thus showing the importance of feed rate control in the machining of the aluminium alloy 6061 while turning, using uncoated carbide insert tool.
- From the ANOVA study, it is found that the depth of cut and cutting speed are insignificant in determining the surface roughness (R_a and R_z) and roundness.
- From the ANOVA study, it is found that the feed rate, depth of cut and cutting speed are insignificant in determining the MRR.
- From the Taguchi's optimization technique, the optimized machining conditions for minimizing the surface roughness (R_a) are found to be a cutting speed of 165.32 m/min, feed rate of 0.05 mm/rev and depth of cut of 0.2 mm with an estimated surface roughness (R_a) of 1.599 µm. The experimental value of surface roughness (R_a) for this combination of parameters is 1.632 µm.
- The percentage of error between the predicted and experimental values of the surface roughness (R_a) during the confirmation experiments is almost within 2%.
- The increase of S/N ratio from the initial process parameters to the optimal process parameters is 1.9996 dB for surface roughness (R_a).
- From the Taguchi's optimization technique, the optimized machining conditions for minimizing the surface roughness (R_z) are found to be a cutting speed of 165.32 m/min, feed rate of 0.05 mm/rev and depth of cut of 0.2 mm with an estimated surface roughness (R_z) of 4.671 µm. The experimental value of surface roughness (R_z) for this combination of parameters is 4.760 µm.
- The percentage of error between the predicted and experimental values of the surface roughness (R_z) during the confirmation experiments is almost within 1.90%.
- The increase of S/N ratio from the initial process parameters to the optimal process parameters is 2.028 dB for surface roughness (R_z).
- From the Taguchi's optimization technique, the optimized machining conditions for minimizing the roundness (Ø) are found to be a cutting speed of 165.32 m/min, feed rate of 0.05 mm/rev and depth of cut of 0.2 mm with an estimated roundness (Ø) of 2.879 µm. The experimental value of roundness (Ø) for this combination of parameters is 2.936 µm.
- The percentage of error between the predicted and experimental values of the roundness (Ø) during the confirmation experiments is almost within 1.97%.
- The increase of S/N ratio from the initial process parameters to the optimal process parameters is 1.931dB for roundness (Ø).
- From the Taguchi's optimization technique, the optimized machining conditions for maximizing the MRR are found to be a cutting speed of 160.45 m/min, feed rate of 0.1 mm/rev and depth of cut of 0.2 mm with an estimated MRR of 0.80742 cm^3/min. The experimental value of MRR for this combination of parameters is 0.7882 cm^3/min.

- The percentage of error between the predicted and experimental values of the response factors during the confirmation experiments is almost within 2.4%
- The increase of S/N ratio from the initial process parameters to the optimal process parameters is 6.2268 dB for Material removal rate (MRR).

The observation of the higher tolerance of the cutting speed clearly indicates the usefulness in increasing the productivity while machining the aluminium alloy 6061, using uncoated carbide insert by operating at higher cutting speeds.

REFERENCES

[1] P.G. Benardos, G.C. Vosniakos, "Predicting surface roughness in machining: a review", *International Journal of Machine Tool and Manufacturing*, Vol.43, 2007, pp. 833-844.

[2] I.P. Arbizu, C.J.L. Perez, "Surface roughness prediction by factorial design of experiments in turning processes", *Journal of Materials Processing Technology*, Vol.143, 2003, pp. 390-396.

[3] M.P. Groover, "Fundamentals of Modern Manufacturing- Materials Processing and Systems", First Ed, Prentice-Hall , Englewood, Cliffs, 1966.

[4] C. Ahilan, Somasundaram kumanan, N. Sivakumaran, J. Edwin Raja Dhas, "Modelling and prediction of machining quality in CNC turning process using intelligent hybrid decision making tools", *Journal of Applied Soft Computing,Vol 25*, 2012, pp. 36-38.

[5] R.Suresh, S. Basavarajappa, G.L. Samuel, "Some studies on hard turning of AISI 41340 steel using multilayer coated carbide tool", *Measurement*, Vol.45, 2012, pp. 1872-1884.

[6] S. Ramesh, L. Karunamoorthy, K. Palanikumar, "Measurement and analysis of surface roughness in turning of aerospace titanium alloy (gr5)", *Measurement* Vol.45, 2012, pp 1266-1276.

[7] N. Muthukrishnan, J. Paulo Davim, "Optimization of maching parameters of Al/Sic-MMC with ANOVA and ANN analysis", *Journal of Materials processing Technology*, Vol. 209, 2009, pp. 225-232.

[8] Chorng-Jyh Tzeng, Yu-Hsin Lin, Yung-Kuang Yang, Ming-Chang Jeng, "Effect of cutting parameters on roughness average, roughness maximum and roundness while turning SKD11 tool steel", *Journal of Materials Processing Technology*, Vol. 209, 2009, pp. 2753-2759.

[9] M. Vijaya kini, A.M. Chincholkar, "Effect of machining parameters on surface roughness and material removal rate in finish turning of glass fiber reinforced polymer pipes", *Materials and Design*, Vol. 31, 2010, pp. 3590-3598.

[10] P. J. Ross, "Taguchi Techniques for Quality Engineering", First Ed, McGraw-Hill, New York, 1996.

[11] K. Palanikumar, "Cutting parameters optimization for surface roughness in machining GFRP composites using Taguchi's method", *Journal of Reinforced Plastic Composites*, Vol.25, 2006, pp. 1739-1751.

[12] Taguchi G., "Introduction to Quality Engineering", Asian Productivity Organization, Tokyo, 1990.